Adaptive Scale Fusion via Uncertainty Estimation for Visual Grounding in Remote Sensing Images

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***Abstract*—Remote sensing visual grounding (RSVG) aims to locate specific objects in remote sensing (RS) images based on referring expressions. While recent methods have achieved promising results by enhancing cross-modal fusion between vision and language, a key challenge remains: significant variation in object scales within a single RS image. Large-scale objects often dominate the representation space, leading to smaller ones being underrepresented or overlooked. To better understand and validate this issue, we analyze typical failure cases and then collect a challenging evaluation set, RSVG-MS, which specifically reflects multiscale discrepancies. Motivated by these insights, we propose ASF-UE, an adaptive scale fusion framework guided by uncertainty estimation. ASF-UE introduces a scale-sensitive attention (SSA) module to extract scale-aware visual features across encoder layers under textual guidance. In addition, a lightweight T-block is designed for spatial alignment, and a spatial uncertainty estimation (SUE) mechanism is integrated to adaptively score and fuse multiscale features. Extensive experiments on two standard benchmarks, OPT-RSVG and DIOR-RSVG, demonstrate the effectiveness of our method. More importantly, ASF-UE achieves significant improvements on the challenging RSVG-MS dataset, highlighting its strength in handling scale variation in remote sensing visual grounding (VG).**

***Index Terms*—Multiscale feature, remote sensing (RS), visual grounding (VG).**

I. Introduction

REMOTE sensing visual grounding (RSVG) imagery [1], [2], [3], [4], [5], [6], [7] aims to accurately identify and localize objects within remote sensing (RS) images [37], [38], [39] based on natural language descriptions. These descriptions [2], [3], [22] typically consist of varied textual phrases or sentences describing attributes such as object category, color, shape, or spatial relationships, with object positions denoted by bounding boxes. This task has gained increasing attention due to its practical significance in enabling nonexperts to intuitively access and interpret vast amounts of RS data [8], [9], including urban planning [5], [8], [40], agriculture monitoring [3], [41], environmental protection [10], [43], and disaster response [11], [12], [42].

Unlike classical referring expression comprehension tasks [8], [13], [14], [25], RSVG presents unique challenges [1], [2], [3], [22] due to the inherent complexity of precisely localizing target objects within RS images. RS images [15], [16] inherently exhibit considerable variability in terms of object scale, occlusion, and spatial arrangement. For instance, large-scale objects like agricultural fields and urban structures coexist with small-scale objects such as airplanes and vehicles. Even within the same category in a single image, as illustrated by ships in Figs. 1 and 2, objects often vary significantly in scale. This scale heterogeneity complicates visual feature extraction [2] and localization [22], making fixed-scale feature extraction [3] and direct attention-based fusion strategies [3], [17] insufficient.

In this task, most existing approaches [1], [2], [3], [22] focus on bridging textual queries and RS images by designing advanced cross-modal fusion frameworks. For instance, GeoVG [1] emphasizes understanding spatial relations in expressions and integrating them with visual features. With the rise of transformer-based models [18], [19], [21], methods like MGVLF [3] and LPVA [2] leverage pretrained backbones for feature extraction and introduce multilevel feature fusion strategies. LPVA further embeds textual features into the visual backbone to enhance feature learning. Recent methods [2], [22], [30] attempt to improve multiscale fusion through cross- and self-attention mechanisms. LPVA reweights visual features using normalized textual embeddings, while LQVG repeatedly applies cross-modal attention over multiscale features before transformer fusion. TACMT upsamples and fuses multiscale visual features via transformers before cross-modal interaction. However, there is a fundamental flaw: visual features at different scales inherently possess distinct spatial semantics and projection ratios, which are not preserved when flattened into uniform sequences for attention. This simplification induces information redundancy and scale-wise competition, ultimately weakening multiscale feature representation [44], [45], [46]. Moreover, despite the use of multilevel fusion, the loss of critical features, especially those of small objects [2], [3], [22], as shown in Fig. 1(a). In Fig. 2, for example, the expression "the ship on the far right" fails to localize the correct small ship, as its feature representation is suppressed during encoding. This limitation hampers accurate reasoning and localization. Therefore, two critical challenges must be addressed: 1) how to effectively capture and leverage multiscale visual features and 2) how to preserve and integrate them adaptively. Fig. 1 illustrates these pitfalls and motivates our proposed scale-sensitive attention (SSA) strategy, further discussed in Section VI-B.

To overcome these limitations, we propose a novel adaptive scale fusion method guided by uncertainty esti-

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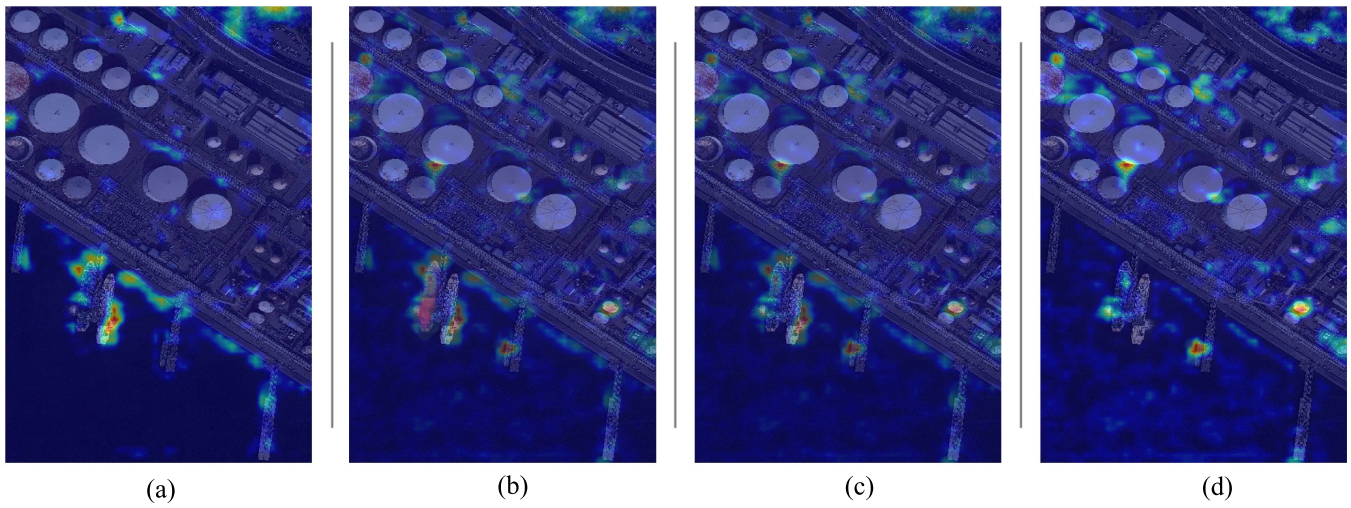


Fig. 1. Motivation behind our ASF-UE approach. In (a), we observe that previous paradigms often fail to localize the correct target when the object scale varies, primarily due to the loss of small objects. (b) Highlights the importance of capturing multilevel visual features for handling objects of various scales. We explicitly preserve multiscale features. In (c), we illustrate that ASF-UE aims to help true small objects receive attention compared to "competing" objects, whereas (d) demonstrates our method's ability to correctly localize the true target using SSA.

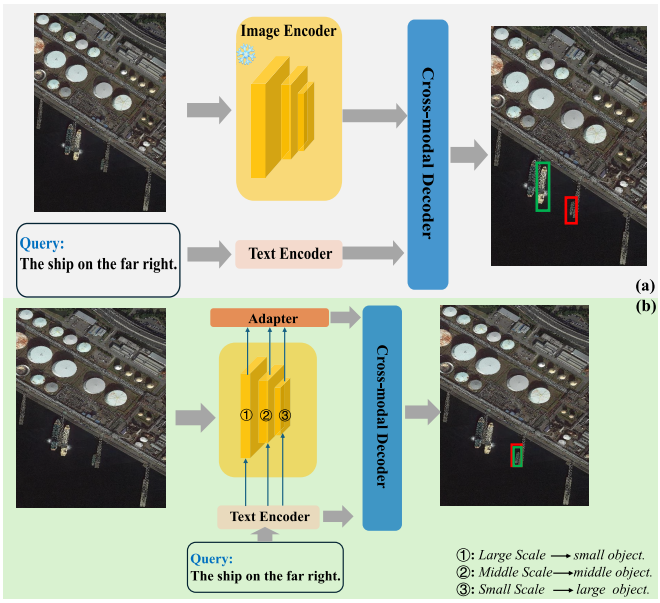


Fig. 2. Comparison between existing methods and our proposed approach for the RSVG task. Green indicates the predicted bounding box; red denotes the ground truth. (a) Most existing methods use a fixed visual backbone to extract features, followed by cross-modal fusion and decoding. (b) In contrast, our ASF-UE framework leverages textual guidance to sensitively capture multiscale object features and adaptively fuse them. By incorporating a UE score, our method reweights scale-specific features, enabling the model to automatically adapt to scale variations.

mation (ASF-UE), explicitly designed to effectively capture multiscale object features and adaptively integrate them, thereby improving cross-modal fusion and target localization. Our approach introduces two primary innovations: the SSA module and the spatial uncertainty estimation (SUE).

1) First, we need to sensitively capture the features of objects referred to by the text across different scales. We introduce a scale-specific embedding token into each layer of the visual encoder, enabling attention-based interactions with the textual features. This design explicitly gives the ability to scale sensitivity into the model. It emphasizes scale-specific differences by maintaining separate, text-guided attention branches for each visual scale. This selective and independent attention preserves intrinsic semantic hierarchies and prevents interscale confusion.
2) After capturing the features at each scale, we need an adapter, the SUE module, which will adaptively integrate multiscale features obtained from SSA. Instead of simple concatenation or self-attention, we designed a T-block for scale projection and an uncertainty computation for each scale features. Then, it computes adaptive scale scores to weigh and merge features across different scales. This adaptive integration ensures precise preservation and fusion of effective targets of varying sizes and complexities. Unlike previous methods [22], [30], which uniformly apply attention mechanisms and neglect distinct projection ratios, our method explicitly encodes each scale's independent projection characteristics. Each scale undergoes independent projection encoding, uncertainty estimation (UE), and adaptive scoring before merging based on computed scale-specific scores, enhancing interpretability and providing clear insights into each scale's relative importance. We also conducted theoretical analysis and experimental validations compared to existing methods.

In summary, our contributions are as follows.

1) We propose a novel adaptive scale fusion framework guided by UE for RSVG. Compared with previous approaches, our framework comprehensively captures multiscale object features and utilizes UE to adaptively and fairly fuse these features into unified visual representations, thereby enabling more precise alignment with referring textual descriptions.

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2) We introduce an SSA module and an SUE mechanism to achieve principled, uncertainty-aware multiscale feature fusion. Compared with standard concatenation or uniform attention, this design preserves semantic granularity and improves fusion effectiveness.
3) We curate and release a refined, challenging dataset by screening over 5500 image–text pairs that explicitly highlight multiscale RSVG scenarios. Experimental analyses demonstrate that this critical validation dataset presents substantial challenges for existing RSVG methods due to pronounced scale variations.
4) Extensive experiments and visualization analyses on the OPT-RSVG and DIOR-RSVG benchmarks,[1] along with our refined multiscale dataset, validate that our approach achieves state-of-the-art (SOTA) performance, confirming the effectiveness and robustness of uncertainty-aware adaptive scale fusion for remote sensing visual grounding (VG).

II. Related Work

A. VG in Remote Sensing Image

VG [13], [14], [15] refers to the task of localizing objects in an image based on a textual description. This classic task has seen significant progress in a two-stage [25], [26], [27] or single-stage framework [18], [19], [20], [28], achieving notable breakthroughs on well-established datasets, such as RefCOCO, RefCOCO+, and Refg. With the advancement of RS [6], [16], [29], VG has been extended to the more challenging domain of RS images. RS data [16], [91] pose more significant challenges due to more pronounced variations in object scales [22], more complex spatial relationships [90], and other practical difficulties. Several studies have explored referring segmentation and detection to advance multimodal understanding in both natural and RS domains [92], [93], [94].

In RSVG, some works have made initial progress. GeoVG [1] addressed these challenges by constructing a geospatial relationship graph to parse textual expressions and localize objects in RS images. MGVLF [3] and LPVA [2] further advanced this task by developing large-scale datasets with more complex images and multigranular textual descriptions to support this task. MGVLF used the last layer of ResNet as the backbone for initial visual feature extraction and added multiconv networks as multiscale visual features, which were then concatenated and enhanced through cross-modal fusion for better scale-specific target extraction. LPVA [2] identified the limitation of fixed visual backbones, which fail to capture objects. They proposed incorporating textual features into the pretrained visual backbone, which helps mitigate the loss of visual information. Building on these previous studies [2], [3], [22], [30], we aim to leverage the multilayer of visual encoders to construct multiscale features. We introduce scale sensitivity [82], [83], [84] to observe and capture the differences between large, medium, and small-scale objects in the visual context [85], [86], guided by textual descriptions. Furthermore, we introduce an adaptive multiscale object fusion strategy [78], [80], [81] instead of concatenation to better handle scale variations in RS images.

[1]https://github.com/johnbager/Adaptive-Scale-Fusion-via-Uncertainty-Estimation-for-Visual-Grounding-in-Remote-Sensing-Images.

B. Multiscale Feature Extraction

Multiscale feature extraction [53], [54], [55], [56], [57] has been widely applied in computer vision to handle varying object sizes [58], ranging from small to large, within one image. In tasks such as object detection [47], [59], [60] and segmentation [63], [85], the pyramidal structures have proved effective by allowing the model to attend to multiple spatial scales without losing contextual information. Early works [47], [64], [65] on multiscale processing often relied on feature pyramids within CNN-based architectures, where features extracted at different network layers were aggregated to capture global context. Feature pyramid networks (FPNs) [47] extended this idea through a top-down architecture with lateral connections [77], [79], enhancing feature maps from higher-level layers and enabling the fusion of low-level details with high-level semantic information. FPN performs predictions at each layer, demonstrating that the features at every scale can contribute to accurate detection. In the field of RS, several studies [50], [51], [52], [62] have also proposed building multiscale object representations. For instance, Li et al. [48] introduced a convolutional pyramid to detect objects across diverse scales in RS target detection, while Li et al. [49] proposed a multiscale contrastive transformation framework. In the realm of RSVG, MGVLF [3] employs a fixed visual encoder to extract base visual features, subsequently adding multiple convolutional layers to form multiscale representations. These representations are then concatenated before cross-modal fusion with textual features. Despite these efforts, existing methods [22], [30] still struggle to balance comprehensive multiscale feature extraction with precise, language-guided object localization in complex RS environments.

C. Adaptive Fusion

The task of VG often requires integrating information from either intramodal [18], [22], [25], [30], [75] or cross-modal [5], [20], [28], [76] sources. One of the main challenges [28], [31], [73], [74] is effectively combining diverse inputs to improve object localization and understanding. Su et al. [31] proposed a gated network to enable language-adaptive reasoning, while some approaches [60], [61], [70], [71], [72], [78] introduced attention-based score distributions for adaptive cross-modal fusion. We introduce an adaptive weighting score mechanism for multiscale visual feature fusion instead of simple concatenation or direct self-attention, thereby guiding multiscale information fusion controllably and explicitly.

III. Preliminaries

A. Multiscale Object in RSVG

In RS imagery, objects often appear at vastly different scales: some are large and salient, while others are small and easily overlooked. Standard feature extraction techniques tend to underrepresent small-scale objects, making their grounding

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particularly challenging, as illustrated in Fig. 2(a). Models are generally less confident when predicting the location of small or ambiguous objects, since the features corresponding to these objects are often too weak compared to larger ones.

B. Uncertainty Estimation

It is the process of quantifying the reliability or confidence of a model's predictions. Cui et al. [87] introduced UE into visual foundation models for open-world object detection, enabling adaptive training for ambiguous and multiscale objects. Durasov et al. [88] applied UE to 3-D object detection, enhancing the model's robustness to missing or out-of-distribution objects. Shao et al. [89] proposed uncertainty-aware fusion to improve small-object detection in autonomous driving scenarios. These methods generally allow the model to adaptively reweight multiple feature components $\{x_1, x_2, \ldots, x_n\}$ under a confidence prior τ. In this article, we incorporate UE to explicitly address multiscale fusion in RSVG. Instead of treating all scales equally, we estimate scale-wise confidence scores that serve as adaptive weights during the fusion process. This probabilistic formulation naturally aligns with the theoretical analysis presented in Section III-C, where uncertainty is modeled as a distribution over scale-specific features.

C. Theoretical Analysis

Compared to conventional top–down architectures, such as FPNs [47], which deterministically aggregate multiscale features in a task-agnostic manner, our approach introduces uncertainty-aware fusion with semantic modulation. Specifically, we view each projected feature map $\tilde{p}_i$ from scale i as a sample drawn from a distribution $p(\mathbf{x}_i \mid \mathcal{L})$, where $\mathcal{L}$ denotes the textual query. We aim to compute the expected aligned feature representation under uncertainty

$$F_v = \mathbb{E}_{p(\mathbf{x}_i|\mathcal{L})}\left[\phi(\mathbf{x}_i)\right] \approx \sum_{i=1}^{N} \alpha_i \cdot \phi(\tilde{p}_i) \cdot \beta_i \tag{1}$$

where $\phi(\cdot)$ is a semantic alignment function, α_i is a weighted score measuring interscale variation, and β_i denotes a spatial confidence mask derived from bidirectional alignment between original and interpolated features. This fusion scheme approximates the posterior predictive distribution, allowing the model to focus on scales that are both text-relevant and spatially coherent.

IV. Method

In this section, we provide a detailed introduction to our final model architecture as shown in Fig. 3 and Algorithm 1, with a particular emphasis on implementing scale sensitivity and adaptive fusion. The following components are discussed: feature encoder, SSA, scale UE, cross-modal decoder, and localization.

A. Feature Encoder

Text Encoder: To ensure a fair comparison [3], [16], [22], we employ the same ResNet [23] as the visual encoder and BERT [24] as the textual encoder for extracting features from input RS images and language expressions. Specifically, given a text expression L composed of l words, we utilize BERT to encode it into textual features $F_l \in R^{L\times C}$. Regardless of variations in visual features [22], [30], our textual features remain unchanged. The textual semantics guide the attention toward visual features. In addition, instead of solely using the [CLS] token as the textual feature representation, we utilize features from all word tokens to achieve multigranular semantic understanding [2], [22].

Image Encoder: For the input RS images, multilevel visual features C_i are extracted using Resnet [23] as a visual backbone. Specifically, as illustrated in Fig. 3, C_3–C_5 correspond to different sizes of H and W, $\{1/8, 1/16, 1/32\}$. We use different stages of downsampled image convolution features [66] as the original multiscale visual features. To maintain a fair comparison with existing RSVG baselines that mainly adopt ResNet-based extractors, we use ResNet-50 as the visual backbone in this work. Nevertheless, our ASF-UE framework is backbone-agnostic and can also be seamlessly integrated with hierarchical vision transformers.

Algorithm 1 ASF-UE: Adaptive Scale Fusion With UE

Require: Remote sensing image I, language query L
Ensure: Predicted bounding box $\hat{b}$

```
Encode L with BERT to obtain text features F_l
Extract multiscale visual features {C_3, C_4, C_5} from visual encoder
for each scale i ∈ {3, 4, 5} do
    Add the scale sensitivity embedding token E_i to C_i
    Compute cross-attention between F_l and C_i to get p_i
end for
for each scale i do
    Project p_i to p̃_i via T-block
    Compute uncertainty map A_i^{lv} and score β_i in eq. 9
    Compute importance score α_i
end for
Fuse features: F_v = Σ_i α_i · p̃_i · β_i
Decode F_v and F_l using FMHCA to obtain F_m
Concatenate F_v, F_l, and reference token p_ref
Pass through Transformer to refine p_ref
Predict bounding box b̂ = MLP(p_ref)
```

B. Scale-Sensitive Attention

To effectively capture and leverage relevant object information across multiple visual scales, we introduce the SSA module. SSA explicitly emphasizes scale-specific differences by maintaining separate text-guided attention branches and sensitive perception for each visual scale. The SSA module operates through the following detailed steps.

To explicitly encode scale-specific characteristics, we introduce a learnable scale embedding token $E_i \in \mathbb{R}^{1\times C}$ for each visual scale C_i. Each embedding token is combined with visual features through token-wise attention and fully connected layers, enabling the model to sensitively perceive scale-specific information

$$\hat{C}_i = \mathrm{FC}\left(\mathrm{Attention}\left(C_i, E_i\right)\right) \tag{2}$$

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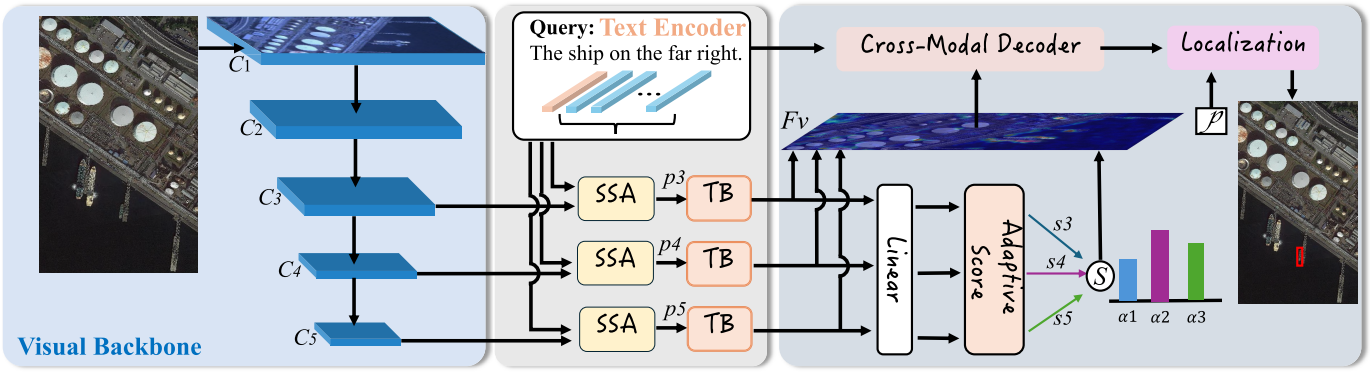


Fig. 3. Overall architecture of our proposed ASF-UE framework. *SSA* computes text-aware multistage visual features c_i, yielding multiscale features p_i. Among these, three nonshared parameters are employed to capture SSA at different scales. *TB* stands for T-Block, the projection weight block to connect SSA and adaptive fusion. An adaptive score will be employed in multiscale features to get the fusion feature F_v, which is computed by UE. Finally, we employ a cross-modal decoder and localization that utilize the text feature along with the fused visual feature F_v to predict the bounding box.

where $\hat{C}_i$ represents scale-aware visual features after the scale-specific embedding. For computational considerations, we use the dot product here for attention calculations. By introducing scale-specific embeddings, SSA encourages the attention layers to operate within distinct representational subspaces for each scale, effectively modeling scale-dependent feature distributions and improving sensitivity to scale variations across objects.

Given textual features $F_l \in \mathbb{R}^{L\times C}$ and scale-aware visual features $\hat{C}_i \in \mathbb{R}^{H_i\times W_i\times C}$, we project these features into queries, keys, and values for attention computation

$$Q_i = F_l W_i^Q, \quad K_i = \text{flatten}\left(\hat{C}_i\right) W_i^K, \quad V_i = \text{flatten}\left(\hat{C}_i\right) W_i^V \tag{3}$$

where W_i^Q, W_i^K, and W_i^V represent linear projection weights for queries, keys, and values at scale i, respectively.

For each visual scale, we independently compute the cross-attention between textual queries and visual keys, obtaining the attention scores as follows:

$$\text{Attn}_i\left(Q_i, K_i, V_i\right) = \text{softmax}\left(\frac{Q_i K_i^\top}{\sqrt{d_k}}\right) V_i \tag{4}$$

where d_k is the dimension of the key vectors, and attention scores quantify how each textual token aligns with different-scale visual regions.

The attention outputs from each scale Attn_i are reshaped back to the original spatial dimensions

$$C_i' = \text{reshape}\left(\text{Attn}_i\left(Q_i, K_i, V_i\right)\right) \in \mathbb{R}^{H_i\times W_i\times C}. \tag{5}$$

This reshaping preserves spatial consistency and retains scale-specific details effectively.

To further emphasize essential visual semantics, we apply a residual connection [23] to fuse the original visual features C_i and the text-guided attention-enhanced features C_i'

$$p_i = C_i + \alpha_i C_i' \tag{6}$$

where α_i is a learnable scaling parameter to control the magnitude of attention enhancement at each scale. This residual aggregation ensures robust integration and mitigates potential information loss.

By performing these steps individually for each scale, our SSA module ensures sensitive and precise alignment of textual semantics with scale-appropriate visual details, preserving distinct intrinsic information at each visual scale. Compared to direct cross-attention, our scale embedding and feature projection can better perceive scale variations.

C. Uncertainty Estimation

While multiscale visual features contain complementary semantic and spatial information, we observe a critical issue in how these features are typically fused [70], [71]. Specifically, due to the varying projection ratios of feature maps at different scales relative to the original image, directly concatenating these and self-attention often introduces a strong scale preference bias. In particular, high-level features like C_5 tend to dominate localization decisions.

To address this issue, we propose an adaptive fusion method with T-block projection and SUE. First, we introduce the T-block to align the spatial and channel dimensions of multiscale features. Second, we estimate scale importance scores to hard integrate the aligned features.

Given SSA visual features $p_i \in \mathbb{R}^{B\times C_i\times H_i\times W_i}$ from scale i, we design a T-block $\mathcal{T}_i(\cdot)$ to project each feature to a unified shape $\mathbb{R}^{B\times C\times H/16\times W/16}$. We use interpolation followed by a 1×1 convolution

$$\tilde{p}_i = \mathcal{T}_i\left(p_i\right) = \text{Conv}\left(\text{Interpolate}\left(p_i\right)\right). \tag{7}$$

The projection ensures all feature maps $\tilde{p}_i$ share the same spatial and channel resolution, allowing downstream fusion. We design a learnable scoring function. Each $\tilde{p}_i$ is first pooled, then passed through two multilayer perceptrons (MLPs)

$$s_i = \text{MLP}\left[\text{AvgPool}\left(\tilde{p}_i\right)\right], \quad \alpha_i = \frac{\exp\left(s_i\right)}{\sum_j \exp\left(s_j\right)} \tag{8}$$

α_i is the fusion weight.

We considered the spatial uncertainty [67], [68], [69]. Directly applying the scale score to all spatial positions may be problematic [22], as some regions are interpolated through T-block projection. We need help correcting the scale redundancy caused by the T-block *fuzzy interpolation area* [88],

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[89]. These interpolated areas, especially in coarser scales, are likely to introduce ambiguity during fusion.

To address this, we design a UE that captures how well the T-block projection preserves the original semantic structure. We further employ a symmetric bidirectional attention design to maintain balanced feature interactions between the original and projected representations, preventing information loss and promoting consistent cross-scale alignment. We propose a symmetric attention formulation to jointly evaluate feature alignment from both directions

$$A_i^{\mathrm{lv}} = \mathrm{softmax}\left(\frac{\left(p_i W_Q\right)\left(\tilde{p}_i W_K\right)^T}{\sqrt{d}} + \frac{\left(\tilde{p}_i W_Q'\right)\left(p_i W_K'\right)^T}{\sqrt{d}}\right). \quad (9)$$

This symmetric formulation measures the bidirectional consistency between the original and projected features at each spatial location. Here, W_Q, W_K, W_Q', and W_K' are learnable linear projections, and A_i^{lv} represents the confidence map. A standard softmax operation is used here to transform raw scores into a probability-like distribution in the projection process. Our symmetric estimation provides a bidirectional and context-aware measure of semantic consistency. This results in more robust UE, particularly under projection-induced distortions, and effectively improves the scale-aware fusion reliability. The reliability score β_i for scale i is then computed by averaging the attention map

$$\beta_i = \mathrm{mean}\left(A_i^{\mathrm{lv}}\right). \quad (10)$$

The scalar β_i acts as a confidence-aware modulation factor. Although β_i is computed via a mean operation, this is applied after SSA has suppressed irrelevant background, making A_i^{lv} already semantically focused. Finally, we incorporate β_i into the multiscale fusion process as

$$F_v = \sum_{i=1}^{3} \alpha_i \cdot \tilde{p}_i \cdot \beta_i. \quad (11)$$

It encourages the model to rely more on projections that better preserve semantic alignment while softly down-weighting unreliable features. The model will adaptively assign weight scores.

D. Cross-Modal Decoder and Localization

After obtaining the adaptively fused multiscale visual representation F_v and the encoded textual features F_l, we integrate the two modalities via a cross-modal decoder [3], [20]. Specifically, the decoder performs multihead cross-attention from the textual features to the fused visual features, enabling the model to refine visual semantics under linguistic guidance

$$F_m = \mathrm{FMHCA}\left(F_v, F_l\right) \quad (12)$$

where F_m denotes the resulting multimodal representation. FMHCA is a transformer-based decoder that decodes visual features conditioned on fine-grained language cues, ensuring effective cross-modal alignment for grounding.

To perform precise localization, we follow the single-token regression paradigm [3], [20], [28], which differs from DETR-style multiquery refinement. Both F_v and F_l are projected into a common embedding dimension and concatenated along with a learnable reference token p_{ref}

$$P = \left[F_v; F_l; p_{\mathrm{ref}}\right]. \quad (13)$$

This token serves as a compact referential query that aggregates contextual information from both modalities. The sequence P is then fed into a transformer encoder with $L = 6$ layers, where multimodal self-attention iteratively updates the representation of p_{ref} to capture the fused referential semantics.

Finally, the localization head is implemented as an MLP with a hidden layer of 256 units and ReLU activation, followed by a linear layer that outputs a 4-D bounding box

$$\hat{b} = \mathrm{MLP}\left(p_{\mathrm{ref}}\right) \in \mathbb{R}^4. \quad (14)$$

This design directly regresses the target bounding box based on the refined referential token, representing a unified visual-language reasoning framework without multistage query decoding. In summary, our cross-modal decoder focuses on linguistic-guided feature alignment, while the localization head performs single-step bounding box prediction from the referential token representation.

Although our decoder follows the single-token regression design for fair comparison with MGVLF, the proposed ASF-UE is flexible and can, in principle, be connected to a DETR-like multiquery refinement decoder, such as LQVG. In this case, the adaptively fused visual representation F_v could serve as the encoder output for multilayer query refinement. However, since SSA already performs text-conditioned fusion in the encoder stage, repeatedly applying cross-modal attention in a DETR-style decoder may introduce semantic redundancy and additional computational overhead.

The strength of our ASF-UE framework lies in its ability to adaptively capture and integrate multiscale features under significant interobject scale variations through uncertainty-guided fusion, effectively mitigating the scale-feature loss problem illustrated in Fig. 1. Benefiting from this adaptive fusion, our framework naturally adheres to a one-stage paradigm that directly predicts the target bounding box from the fused visual features conditioned on the textual query.

V. Experiments

In this section, we conduct extensive experiments to evaluate the effectiveness of our proposed ASF-UE framework for RSVG. We organize our evaluation across five subsections. In Section V-A, we provide comprehensive implementation details of our method. Sections V-B and V-C introduce the benchmark datasets and evaluation metrics used. In Section V-D, we compare our approach against several SOTA methods. Section V-E presents a series of ablation studies to analyze the contribution of each module. Finally, Section VI offers qualitative visualizations to illustrate the interpretability and effectiveness of our model.

A. Implementation Details

We set the input image size as 640×640. We set the maximum length of languages to 40. We adopt ResNet-50 and

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BERT as the backbone networks for visual and textual encoding, respectively. Specifically, we use the ResNet-50 weights initialized from the DETR [36] model. For the text encoder, we utilize BERT [24] with a hidden size of 768 to encode the referring expressions. We also apply the same image resize and data augmentation strategies [20], including random cropping, scaling, and translation. The model is optimized using the AdamW optimizer with a weight decay of 10^{-4}. The learning rate is set to 1×10^{-5}. Training is conducted for 150 epochs with a batch size of 16. All experiments are run on four 3090 GPUs.

B. Datasets

DIOR-RSVG and OPT-RSVG: We evaluate the effectiveness of our method on two large-scale remote sensing VG benchmarks. The first is the DIOR-RSVG dataset, which is constructed based on the RS image DIOR dataset [16]. DIOR-RSVG consists of 38 320 language expressions and 17 402 RS images. Each image is sized at 800 × 800 pixels, depicting 20 object categories. The average length of each expression is 7.47 words. These concise expressions primarily describe the target object's category and a small amount of spatial information, making DIOR-RSVG a challenging yet representative benchmark for evaluating grounding accuracy in RS imagery. The second, OPT-RSVG, is constructed from three RS object detection datasets (HRRSD, DIOR, and SPCD). It contains 25 452 RS images and 48 952 image–query pairs spanning 14 object categories. Each image is accompanied by a language expression describing the target object, which yields an average referring expression length of 10.10 words (ranging from 3 to 32). By encompassing a wide spectrum of target scales (from tiny to large), images captured from diverse viewing angles, variations in color shading due to different sensors, and complex background noise, OPT-RSVG presents a demanding testbed for remote sensing VG.

Statistics of Multiscale Variation (RSVG-MS): As shown in Table I, we constructed a specialized dataset, named RSVG-MS, designed explicitly to evaluate the multiscale object variation problem prevalent in remote sensing VG. This dataset was derived through a systematic selection and reannotation process from two widely-used large-scale RSVG datasets: DIOR-RSVG and OPT-RSVG. Specifically, we first manually categorized and screened all available images from these datasets, totaling 42 854 RS images. Through initial screening, we identified 19 139 images (denoted as MS0) containing evident multiscale variation issues. Among these, 9964 images (MS01) displayed multiscale variations within objects of the same category, while 9175 images (MS02) exhibited scale differences across different object categories. Subsequently, we conducted a further manual reannotation and filtering step, aiming to select images with clear and pronounced multiscale discrepancies within the same category. After this rigorous second screening, we produced a refined subset consisting of 5505 images clearly highlighting the multiscale variation challenge.

The final RSVG-MS dataset thus provides a robust benchmark explicitly designed to evaluate models' capabilities in handling multiscale object variations in remote sensing VG tasks.

TABLE I
STATISTICS OF THE CONSTRUCTED RSVG-MS DATASET. RESULTS SHOW THAT RSVG-MS ACCOUNTS FOR 12.85% OF THE RSVG DATASETS

Dataset Stage	Number of Images
Total images from DIOR-RSVG and OPT-RSVG	42,854
Images containing multi-scale (MS0)	19,139
- Within-category multi-scale (MS01)	9,964
- Cross-category multi-scale (MS02)	9,175
Final refined subset	5,505

C. Metrics

Following [3], we report Pr@{0.5, 0.6, 0.7, 0.8, 0.9}, meanIoU, and cumIoU as our primary evaluation metrics. MeanIoU and CumIoU, formulated as

$$\text{meanIoU} = \frac{1}{M}\sum_{t=1}^{M}\frac{I_t}{U_t}, \quad \text{cumIoU} = \frac{\sum_{t=1}^{M} I_t}{\sum_{t=1}^{M} U_t} \tag{15}$$

where M is the total number of test samples, and I_t and U_t are the intersection and union areas, respectively, between the predicted and ground-truth bounding boxes for the tth sample. These metrics collectively provide a comprehensive evaluation of our model's localization accuracy across various IoU thresholds.

D. Comparisons With SOTA Methods

As shown in Tables II and III, our method achieves the best performance on both the DIOR-RSVG and OPT-RSVG benchmarks, consistently surpassing other SOTA approaches in terms of Pr@0.5–0.9, meanIoU, and cumIoU. We also conducted comparative experiments to evaluate how existing methods perform on the multiscale variation problem using our constructed RSVG-MS dataset.

Early works such as ZSGNet [35], FAOA [19], ReSC [18], and LBYL-Net [34] are classic one-stage frameworks for visual localization, without transformer mechanisms. Recently, Transformer-based approaches, e.g., TransVG [20], MGVLF [3], and LPVA [2], have advanced RSVG performance by employing cross-modal attention. As our motivation indicated in the introduction, in addition to incurring higher computational overhead, treating multiscale features as sequence can lead to semantic bias and confusion. ASF-UE directly addresses these issues by applying SSA independently at each scale to preserve fine-grained cues for small objects and coarse semantics for larger ones, introducing a T-block projection with adaptive uncertainty to align and filter multiscale features, and designing an adaptive fusion mechanism based on UE that explicitly models scale-specific fusion order. Owing to this design, ASF-UE outperforms prior SOTA methods. On DIOR-RSVG, it registers a significant improvement in meanIoU and cmuIoU, demonstrating a strong ability to handle diverse scales and reduce referential ambiguity. On the more challenging OPT-RSVG benchmark with greater scale diversity and

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TABLE II

COMPARISON WITH SOTA METHODS ON THE DIOR-RSVG TEST SET. BOLD FONT INDICATES THE BEST PERFORMANCE AMONG ALL METHODS. UNDERLINED VALUES DENOTE THE BEST PERFORMANCE AMONG PUBLISHED BASELINES

Methods	Venue	Visual Encoder	Language Encoder	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	meanIoU	cmuIoU
ZSGNet [35]	ICCV'19	ResNet-50	BiLSTM	51.67	48.13	42.30	32.41	10.15	44.12	51.65
FAOA [19]	ICCV'19	DarkNet-53	BERT	67.21	64.18	59.23	50.87	34.44	59.76	63.14
ReSC [18]	ECCV'20	DarkNet-53	BERT	72.71	68.92	63.01	53.70	33.37	64.24	68.10
LBYL-Net [34]	CVPR'21	DarkNet-53	BERT	73.78	69.22	65.56	47.89	15.69	65.92	76.37
Transformer-based:										
TransVG [20]	CVPR'21	ResNet-50	BERT	72.41	67.38	60.05	49.10	27.84	63.56	76.27
VLTVG [33]	CVPR'22	ResNet-50	BERT	69.41	65.16	58.44	46.56	24.37	59.96	71.97
MGVLF [3]	TGRS'23	ResNet-50	BERT	76.78	72.68	66.74	56.42	35.07	68.04	78.41
LPVA [2]	TGRS'24	ResNet-50	BERT	82.27	77.44	72.25	60.98	39.55	72.35	85.11
LQVG [22]	TGRS'24	ResNet-50	BERT	83.41	81.03	75.91	65.52	43.53	74.02	82.22
GLAED [90]	TGRS'25	ResNet-50	BERT	84.44	81.52	73.51	62.92	43.36	73.69	82.92
ours	-	ResNet-50	BERT	**85.73**	**81.64**	**76.87**	**66.13**	**44.01**	**75.31**	**85.92**

TABLE III

COMPARISON WITH THE SOTA METHODS ON THE TEST SET OF OPT-RSVG. BOLD FONT INDICATES THE BEST PERFORMANCE AMONG ALL METHODS. UNDERLINED VALUES DENOTE THE BEST PERFORMANCE AMONG PUBLISHED BASELINES

Methods	Venue	Visual Encoder	Language Encoder	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	meanIoU	cmuIoU
ZSGNet [35]	ICCV'19	ResNet-50	BiLSTM	48.64	47.32	43.85	27.69	6.33	43.01	47.71
FAOA [19]	ICCV'19	DarkNet-53	BERT	68.13	64.30	57.15	41.83	15.33	58.79	65.20
ReSC [18]	ECCV'20	DarkNet-53	BERT	69.12	64.63	58.20	43.01	14.85	60.18	65.84
LBYL-Net [34]	CVPR'21	DarkNet-53	BERT	70.22	65.39	58.65	37.54	9.46	60.57	70.28
Transformer-based:										
TransVG [20]	CVPR'21	ResNet-50	BERT	69.96	64.17	54.68	38.01	12.75	59.80	69.31
VLTVG [33]	CVPR'22	ResNet-50	BERT	71.84	66.54	57.79	41.63	14.62	60.78	70.69
MGVLF [3]	TGRS'23	ResNet-50	BERT	72.19	66.86	58.02	42.51	15.30	61.51	71.80
LPVA [2]	TGRS'24	ResNet-50	BERT	78.03	73.32	62.22	49.60	25.61	66.20	76.30
LQVG [22]	TGRS'24	ResNet-50	BERT	78.58	75.26	**69.03**	**55.21**	26.83	67.25	75.59
ours	-	ResNet-50	BERT	**82.34**	**76.31**	66.17	52.41	**27.93**	**70.03**	**79.49**

TABLE IV

COMPARISON ON THE RSVG-MS DATASET. ALL MODELS ARE TRAINED ON THE ORIGINAL DATASETS AND TESTED ON RSVG-MS WITHOUT FINE-TUNING. * INDICATES THAT ALL MODELS WERE TRAINED AND TESTED UNDER THE SAME DATASET AND ENVIRONMENT SETTINGS

Method	Pr@0.5	MeanIoU	cmuIoU
ReSC*	34.91	28.21	29.34
TransVG*	36.46	27.78	29.14
LPVA*	41.21	30.17	33.26
LQVG*	44.73	32.10	33.82
Ours	**59.10**	**43.42**	**48.83**

TABLE V

ABLATION STUDY OF THE SSA AND ADAPTER MODULE

	SSA	Adapter	Pr@0.5
①	last layer+	Adapter	72.75
②	Multi-layer+	Concat	75.51
③	Multi-layer+	Adapter	81.17
④	SSA	+Concat	81.97
⑤	SSA	+Weight Concat	83.11
⑥	SSA	+Self-attention	85.21
	SSA	SUE	**85.73**

complex backgrounds, ASF-UE also achieves new top scores, affirming its robustness and generalization capability in remote sensing VG.

We tested all models, trained on the original OPT-RSVG and DIOR-RSVG datasets, directly on RSVG-MS without any further fine-tuning, keeping their parameters frozen. This setting allows us to assess the ability of existing approaches to handle scale variation. As shown in Table IV, our method significantly outperforms all baselines under this evaluation protocol. This result highlights the effectiveness of our approach in addressing the challenges posed by multiscale variation in remote sensing VG.

E. Ablation Study

We perform ablation experiments on the DIOR-RSVG dataset to evaluate the contributions of our SSA module and *adaptive score fusion* and others. We also present qualitative visualizations to further validate the effectiveness of these modules in Fig. 1. To improve interpretability and provide a clearer understanding of the contribution of each module, we reorganized the ablation results into two dedicated tables. Table V focuses on the main proposed modules that form the core of our framework, while Table VI summarizes the auxiliary components, such as the scale-specific embedding, the T-block, and the UE. This organization allows a more transparent analysis of how each module individually and collectively contributes to the overall performance.

1) SSA: In the first three rows of Table V, we design three groups of comparative experiments to verify multiscale features and the SSA mechanism. In ①, we disable multiscale processing by only retaining the final scale of the visual backbone as the feature output, which is also shown in

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TABLE VI
ABLATION STUDIES OF PROPOSED MODULES

(a)			(b)			(c)			(d)	
Specific Embedding	Pr@0.5	meanIoU	Method	Pr@0.5	meanIoU	Method	Pr@0.5	meanIoU	Method	RunTime (s)
w/o SE	83.07	73.98	T-block Interpolation	**85.73**	75.31	MLP-based	83.11	73.21	MGVLF	1.1
w/ SE	**85.73**	**75.31**	Transpose Conv.	85.08	**75.39**	UE	**85.73**	**75.31**	Ours	1.5

Fig. 1(a). In ② and ③, we retain multiple feature scales, but feed them directly into subsequent computations without the proposed SSA. Experimental results confirm that performing cross-modal attention *at each scale* before fusion is beneficial, aligning with prior findings [2], [3] that emphasize the value of multiscale representation. Compared to ②, which is a baseline for multiscale VG, it shows our method's strong performance.

2) Adaptive Fusion: In the last three rows of Table V, we further construct three comparative experiments to assess our *adaptive score fusion* design. First, we replace both T-block and UE with a simple *concat* operation, in ② and ④. Next, we substitute the T-block projection with an MLP-based weighting to produce multiscale weights and then weighted concatenation, indicated as "SSA + Weight concat." At last, we apply a straightforward self-attention mechanism in place of our adaptive score fusion. Across all comparisons, our method shows that explicitly modeling scale-wise projection and reliability indeed enhances multiscale fusion. Compared to ⑥, we validate our motivation claim that directly treating multiscale features as a single sequence for self-attention neglects crucial scale differences.

3) Scale-Specific Embedding: As shown in 2, we add a scale-specific embedding layer to the visual encoder. This embedding introduces scale sensitivity into attention computation without modifying the original visual features. Table VI(a) demonstrates its effectiveness within SSA. w/o SE means that we only cancel the connection to ensure postorder training.

4) T-Block Projection: We compare interpolation and transpose convolution for feature alignment, as shown in Table VI(b). Compared to traditional methods like transpose convolution, which suffer from sparse updates and overlapping artifacts, our T-block combines interpolation with a 1×1 convolution to preserve semantic structure in a lightweight manner. This design also facilitates integration with our UE module, enabling more reliable multiscale fusion under projection distortion.

5) Uncertainty Estimation: We compute a spatial uncertainty confidence map per scale. It is obtained through bidirectional attention between the preprojection and postprojection features (p and $\tilde{p}$). Table VI(c) compares our approach with an MLP-based alternative. As shown in (d), the inference speed remains within an acceptable range despite the performance improvement.

VI. VISUALIZATION

A. Visualization Results

As shown in Fig. 4, we present the visualization results comparing our method with previous approaches [3], alongside the ground truth. It is evident that our method significantly outperforms the existing methods in handling multiple objects with large-scale variation within the same image, which often leads to localization errors. This demonstrates the effectiveness of our proposed SSA and adaptive score in addressing this issue. Specifically, compared to other methods that employ multiscale features but with simpler approaches, such as direct concatenation or self-attention mechanisms, our method more effectively handles scale discrepancies, avoiding localization failures that occur when such targets are present.

We also show the failure cases of our method in Fig. 5. Fig. 5(a) indicates that our method still exhibits shortcomings in high-level semantic understanding. Going forward, we plan to incorporate more advanced large-scale models, such as comprehensive vision-language models, into the cross-modal decoding and localization module to further enhance reasoning capabilities.

B. Attention Visualization

As shown in Fig. 1, we also present the heatmap results to show our ASF-UE motivation. In detail, we found the small-scale object grounding error in Fig. 1(a). We hypothesize that, within the same image, targets of the same category can vary greatly in scale, causing existing methods to overlook small-scale objects and ultimately fail at grounding. Therefore, we explicitly preserve multiscale features, as depicted in Fig. 1(b), and use them in subsequent computations. As shown in Fig. 1(c), this approach enables genuinely small objects to receive more attention compared to "competing" targets. Finally, by adaptively fusing these features, we arrive at the correct localization results, as illustrated in Fig. 1(d).

VII. DISCUSSION

Recently, an increasing number of vision-language large models (VLLMs) have been explored for remote sensing VG. These models [5], [40] possess strong general visual-language alignment capabilities and knowledge acquired through large-scale pretraining. Nevertheless, directly applying general-purpose VLLMs to RS imagery remains challenging due to the unique characteristics of RS data, such as large intraobject and interobject scale variations and complex spatial distributions. In this context, our proposed ASF-UE framework can serve as a lightweight and complementary module to enhance VLLM-based pipelines. Specifically, the SSA and SUE modules can be seamlessly integrated into the visual encoder of a VLLM to provide uncertainty-guided, scale-adaptive feature representations, which may help alleviate the recognition difficulties caused by large object-scale variations

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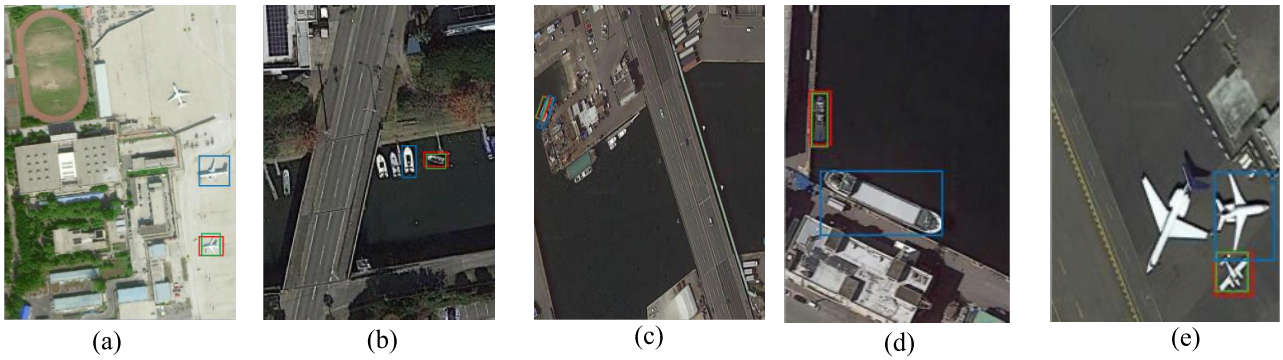


Fig. 4. Visualization of experimental results. The red bounding box represents the ground truth, the green one shows the result from our method, and the blue one is the prediction from the previous method [3]. (a) Small plane below. (b) Ship on the far right. (c) Top left ship in the top ships. (d) Boat on top. (e) Target to locate is the small plane in the bottom right.

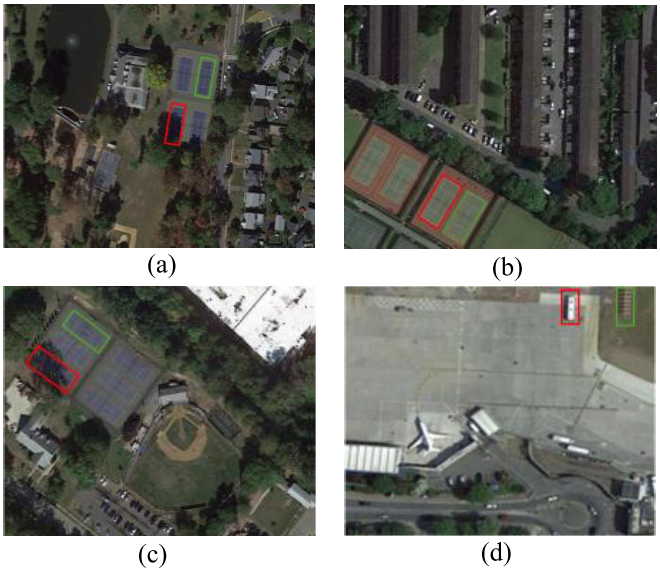


Fig. 5. Failure cases of our method. These instances highlight challenging targets that require high-level semantic understanding. (a) Basketball court is at the bottom left, and the target is the tennis court at the top right. (b) Tennis court to the right of the basketball court. (c) Tennis court in the top left. (d) Truck in the top right corner.

commonly observed in RS imagery. We believe this integration offers a promising direction for improving the localization accuracy and robustness of large-model-driven VG in remote sensing scenarios.

VIII. Conclusion

In conclusion, we present a novel adaptive scale fusion method via UE to address the challenge of large-scale variability in remote sensing VG. By employing an SSA module for language-guided multiscale association, we effectively capture objects of diverse sizes. We further introduce a T-block projection and UE to mitigate the ambiguity and adaptively fuse these multiscale features. Through this design, our method achieves SOTA results in both qualitative and quantitative experiments, demonstrating the efficacy of scale-sensitive fusion for robust localization in RS imagery. Finally, we also present failure cases and discuss potential directions for future research.

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